

CONTACT INFORMATION

University of California, Los Angeles
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EMPLOYMENT

University of California, Los Angeles, Los Angeles, CA, USA

September 2026 – present

Inaugural Bhaumik Prize Postdoctoral Fellow
Supervisor: Anshul Kogar

National High Magnetic Field Laboratory & Florida State University, Tallahassee, FL, USA

September 2024 – August 2026

Quantum Initiative Postdoctoral Fellow
Supervisor: Cyprian Lewandowski

EDUCATION

The Ohio State University, Columbus, OH, USA

August 2019 – September 2024

Ph.D. in Physics
Advisor: Brian Skinner

Indian Institute of Science Education and Research, Pune (IISER Pune), India

August 2014 – May 2019

Integrated BS-MS *with Distinction* (Major - Physics)
Master Thesis Advisor: Sreejith G J

RESEARCH INTERESTS

Theoretical modeling of correlated quantum materials motivated by experimental observations; focusing on emergent quantum phases and phase transitions driven by Coulomb interaction, chirality, quantum geometry, and disorder.

PUBLICATIONS

• Theoretical Works

1. Sandeep Joy, Brian Skinner, Disorder-induced liquid-solid phase coexistence in 2D electron systems, *Phys. Rev. B (Letter)* **113**, L201117 (2026), *Editors' Suggestion*
2. Mohammed Hammam, Cyprian Lewandowski, Vladimir Dobrosavljevic, Sandeep Joy, Is disorder a friend or a foe to melting of Wigner-Mott insulators?, [arXiv:2512.07932](https://arxiv.org/abs/2512.07932) (2025)
3. Sandeep Joy, Leonid Levitov, Brian Skinner, Chiral Wigner crystal phases induced by Berry curvature, *Phys. Rev. Lett.* **135**, 256502 (2025)
4. Sandeep Joy, Brian Skinner, Wigner crystallization in Bernal bilayer graphene, [arXiv:2310.07751](https://arxiv.org/abs/2310.07751) (2023)
5. Sandeep Joy, Brian Skinner, Upper bound on the window of density occupied by microemulsion phases in two-dimensional electron systems, *Phys. Rev. B (Letter)* **108**, L241110 (2023)
2. Sandeep Joy, and Brian Skinner, Wigner crystallization at large fine structure constant, *Phys. Rev. B (Letter)* **106**, L041402 (2022)
1. Sandeep Joy, Saad Khalid, and Brian Skinner, Transparent mirror effect in twist-angle-disordered bilayer graphene, *Phys. Rev. Research* **2**, 043416 (2020)

• Joint Experiment-Theory Works

3. Zekang Zhou, Kilian Kröttsch, Raphaël Ayache, Yonggen Li, Sandeep Joy, Kenji Watanabe, Takashi Taniguchi, Moty Heiblum, Preden Rouleau, Mitali Banerjee, Competing Orders Driven by Wigner Crystal Phase in Rhombohedral Graphene, [arXiv:2607.15014](https://arxiv.org/abs/2607.15014) (2026) (under review in *Nature Physics*)
2. Zhenqi Hua[†], Chang Niu[†], Sandeep Joy[†], Pukun Tan, Gang Shi, Haoyang Liu, Jiaying Guo, David Graf, Peide Ye, Cyprian Lewandowski, Peng Xiong, Interplay of orbital and spin magnetization in trigonal tellurium, [arXiv:2507.14292](https://arxiv.org/abs/2507.14292) (2025) ([†]Z.H., C.N., and S.J. contributed equally to this work, under review in *Phys. Rev. Lett.*)

1. Fangyuan Yang, Ruiheng Bai, Alexander A. Zibrov, Sandeep Joy, Takashi Taniguchi, Kenji Watanabe, Brian Skinner, Mark O. Goerbig, Andrea F. Young, Cascade of multi-electron bubble phases in monolayer graphene at high Landau level filling, Phys. Rev. Lett. 131 (2023), *Editors' Suggestion*

• **Commentaries/Reviews/Perspective**

1. Watching electronic ice melt, Science 388 (2025) (commentary on Z. Xiang et al., Science 388 (2025).)

**SELECTED
PROFESSIONAL
SERVICE**

- Journal Referee Services: Reviewer for *Science*, *Physical Review Letters*, *Physical Review B*, *Physical Review Applied*, *Newton (Cell Press)* and *Phase Transitions*
- Reviewer for *French National Research Agency (ANR) grant proposal*
- Reviewer for *Postdoctoral Scholars Career Development Travel Award by Florida State University Provost's Office and the Office of Postdoctoral Affairs*, 2026
- Session Chair, *Advances in Numerical Modeling of Quantum Phase Transitions, Global Physics Summit*, 2026
- Sorter for Division of Condensed Matter Physics sessions on *APS Global physics summit*, 2026
- Session Chair, *New Horizons in Quantum Hall Effects, Chern Insulators, and Superconductivity: Theory Winter School*, National High Magnetic Field Laboratory 2026

**SELECTED
FELLOWSHIPS**

- Inaugural Bhaumik Prize Postdoctoral Fellowship, University of California Los Angeles, 2026 - present
- Quantum Initiative Postdoctoral Fellowship, Florida State University/National High Magnetic Field Laboratory, 2024 - 2026
- Junior Research Fellowship by Council of Scientific and Industrial Research, Government of India, 2018
- German Academic Exchange - Working Internship in Science and Engineering (DAAD-WISE) Fellowship, 2017
- Indian Academy of Sciences Summer Research Fellowship, 2016
- KVPY Fellowship, Department of Science and Technology, Government of India, August 2015 - July 2019
- INSPIRE Fellowship, Department of Science and Technology, Government of India, August 2014 - July 2019 (discontinued to accept KVPY)

**SELECTED TRAVEL
AWARDS**

- Postdoctoral Scholars Career Development Travel Award by Florida State University Provost's Office and the Office of Postdoctoral Affairs, 2026
- Distinguished Student (DS) Travel Grant by American Physical Society Forum on International Physics and Forum for Early Career Scientists, 2026
- Edward J. Ray Travel Award for Scholarship and Service, The Ohio State University, 2021

SELECTED GRANTS

- Undergraduate Research Opportunity Program Research Mentor Materials Grant funding, Florida State University, 2025

INVITED TALKS

Condensed Matter Physics: “reshaping” universe at will

- Research Experiences for Teachers (RET), National High Magnetic Field Laboratory, 2026

Disorder: friend or foe to the melting of 2D electron solid? Two vignettes

- Gordon Research Seminar on Emergent Properties and Topology in Quantum Materials and Engineered Systems, 2026
- Florida Quantum Conference, Florida State University, 2026

Chiral Wigner Crystal Phases Induced by Berry Curvature

- Invited Session on Topological Electron Crystals, Global Physics Summit, 2026

The story of Wigner crystallization in Bernal bilayer graphene

- Condensed Matter Sciences Seminars, National High Magnetic Field Laboratory, 2024

Transparent mirror effect in twist-angle-disordered bilayer graphene

- Indian Institute of Science Education and Research, Pune, India, 2020

CONTRIBUTED TALKS

Is disorder a friend or a foe to melting of Wigner-Mott insulators?

- APS March Meeting, American Physical Society, 2026

Disorder-induced liquid-solid phase coexistence in 2D electron systems

- APS March Meeting, American Physical Society, 2025

How prominent are microemulsion phases in 2D electron systems?

- APS March Meeting, American Physical Society, 2024

Wigner crystallization in Bernal bilayer graphene

- APS March Meeting, American Physical Society, 2023

Wigner crystallization at large fine structure constant

- APS March Meeting, American Physical Society, 2022

Transparent mirror effect in twist-angle-disordered bilayer graphene

- International Conference on Low Energy Electrodynamics in Solids (LEES), 2021
- Hayes Graduate Research Forum, The Ohio State University, 2021
- APS March Meeting, American Physical Society, 2021

POSTER PRESENTATIONS

Disorder: friend or foe to the melting of 2D electron solid? Two vignettes

- Quantum Simulation of Strongly Correlated Quantum Matter, CTEQ Summer School, Penn State University, 2026
- Gordon Research Conference on Correlated Electron Systems: Collective Phenomena, Entanglement and Topology in Quantum Materials, 2026

The nature of the quantum liquid-solid transition for 2D electrons

- Boulder School 2025: Dynamics of Strongly Correlated Electrons
- FSU Quantum Initiative - Dirac Quantum Discussions Symposium, Florida, 2025
- Strong Correlation Across Newly Accessible Length and Energy Scales, Theory Winter School, National High Magnetic Field Laboratory, 2025

How prominent are microemulsion phases in 2D electron systems?

- Gordon Research Conference on Correlated Electron Systems: Unconventional Phenomena in Quantum Matter, Massachusetts, 2024

Wigner crystallization in Bernal bilayer graphene

- Q-PHORIA: Quantum Pennsylvania Ohio Regional Annual Conference, University of Pittsburgh, 2023
- Novel Quantum States of Matter in Moiré Materials, Aspen Center for Physics - Winter Conference, 2023
- Correlations in Flat Bands: From the FQHE to Moiré, Theory Winter School, National High Magnetic Field Laboratory, 2023
- Strongly Correlated Matter: from Quantum Criticality to Flat Bands, International Center for Theoretical Physics, Italy, 2022
- Gordon Research Conference on Correlated Electron Systems: Topology and Correlations: Long-Range Entanglement in Many-Body Systems, 2022

COMPUTATIONAL SKILLS

- **Computational Languages & software stacks**
Python, Mathematica, Matlab, Qiskit and L^AT_EX
- **Numerical Skills**
Transfer Matrix Methods, Exact Diagonalization, Variational wave functional methods, Numerically solving ODE's, Dynamical Mean Field Theory, Phenomenological modeling, Machine Learning

LICENSES & CERTIFICATIONS

1. Erdős Institute Summer 2025 Quantum Computing Bootcamp
2. Institute of Physics Peer Review Excellence online training graduate

MENTORING EXPERIENCE

- Gordon research conference mentor for first time attendees (2026)
- Graduate Students: Joshua Scales (OSU), Reilly McDowell (OSU), Mohammed Hammam (FSU/MagLab)
- Undergraduate Students: Reese Boucher (West Virginia University, OSU CEM Summer REU Program, 2020), Luke Knull (OSU Polaris Mentorship Program , 2021-2022), Ramon Morales III (FSU UROP Program, 2025-2026), Mira Best (Embry-Riddle Aeronautical University, MagLab summer REU program, 2026)
- High school students: Kai Okui (MagLab High School Externship, 2025-2026)

TEACHING EXPERIENCE

Substitute Instructor, Florida State University

- Phase Transitions and Critical Phenomena (Spring 2026), Course instructor: Vladimir Dobrosavljević

Graduate Teaching Assistant, The Ohio State University

- Physics 5501H - Honors Quantum Mechanics 2 and Physics 5401H Honors Advanced Electricity and Magnetism 2, Spring 2021
- Physics 5500H - Honors Quantum Mechanics 1, Fall 2020
- Physics 5400 - Intermediate Electricity and Magnetism, Spring 2020
- Physics 1250 - Mechanics, Work and Energy, Thermal Physics, Fall 2019

OUTREACH ACTIVITIES

- Co-organizer, *FSU Quantum Discovery Day 2025*
- *Chief* quantum ambassador, [FSU Quantum Initiative](#), 2025 - present
- Organizer and founder *Quantum Matter Journal Club*, Condensed Matter Science Division, National High Magnetic Field Laboratory, 2025 - present
- Judge at Capital Regional Science and Engineering Fair, Tallahassee, Florida for middle school students, 2025
- Volunteer, MagLab (National High Magnetic Field Laboratory) representative at local school STEM events, 2024–present
- Trained tour guide at the MagLab (National High Magnetic Field Laboratory), 2024 - present
- Founding member and treasurer of graduate student organization “Random Interactions,” 2022-2024 Aims to foster engagement and collaboration among undergraduate students, graduate students, postdocs, and faculty working in condensed matter physics.
- Judge at State Science Day for high school and middle school students at Ohio Science Academy, 2021-2024.
- Elected Graduate Studies Committee student representative, Physics Department, 2021 - 2022

REFERENCES

1. Brian Skinner

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